

Online supplement

Carotid Artery Intima-Media Thickness and Distensibility in Children and Adolescents: Reference Values and Role of Body Dimensions

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- **Subject Characteristics for the Reference Population and Hypertensive/Obese Subjects**
- **Percentile and LMS Tables for cIMT, Einc, beta and DC**
- **Comparison of Height and Age Normalization**
- **Equations for Blood Pressure and Distensibility Calculations**

Table S1: Subject characteristics of the reference population (n=1051) and obese or hypertensive participants (n=104). m=male, f=female

Parameters	Reference population	Obese/Hypertensive participants
No. of Subjects		
m	492	67
f	559	37
Age (years)		
m	12.5 ± 3.36	10.7 ± 2.24
f	12.1 ± 3.0	11.0 ± 2.1
Height (cm)		
m	154.9 ± 20.7	150.0 ± 17.1
f	150.2 ± 15.1	150.6 ± 11.6
Height SDS		
m	0.40 ± 1.10	0.98 ± 1.12
f	0.33 ± 1.0	0.97 ± 1.29
BMI (kg/m ²)		
m	19.1 ± 3.21	24.6 ± 2.89
f	19.1 ± 3.00	26.0 ± 3.07
BMI SDS		
m	0.26 ± 0.98	2.47 ± 0.33
f	0.23 ± 0.95	2.41 ± 0.37
Systolic BP (mmHg)		
m	110.8 ± 12.0	110.3 ± 9.4
f	107.3 ± 9.8	113.6 ± 10.0
Systolic BP SDS		
m	0.15 ± 0.88	0.33 ± 0.74
f	0.11 ± 0.89	0.84 ± 0.9
Diastolic BP (mmHg)		
m	59.0 ± 7.1	60.8 ± 6.4
f	59.2 ± 7.4	63.3 ± 7.2
Diastolic BP SDS		
m	-0.34 ± 0.59	-0.19 ± 0.65
f	0.27 ± 0.64	0.00 ± 0.67

Table S2: Percentile tables for cIMT for boys and girls.

<i>cIMT</i>														
Age and Height	<i>BOYS</i>							<i>GIRLS</i>						
	P5	P10	P25	P50	P75	P90	P95	P5	P10	P25	P50	P75	P90	P95
Age (years)														
6	0.2921	0.3100	0.3388	0.3695	0.3992	0.4251	0.4403	0.297	0.313	0.338	0.366	0.394	0.418	0.432
6.5	0.2938	0.3114	0.3400	0.3707	0.4005	0.4266	0.4420	0.299	0.315	0.340	0.368	0.395	0.419	0.433
7	0.2954	0.3128	0.3412	0.3719	0.4018	0.4282	0.4437	0.301	0.317	0.342	0.369	0.396	0.420	0.435
7.5	0.2971	0.3142	0.3424	0.3730	0.4031	0.4297	0.4454	0.303	0.319	0.344	0.371	0.398	0.422	0.436
8	0.2988	0.3157	0.3436	0.3741	0.4043	0.4311	0.4471	0.305	0.320	0.345	0.373	0.399	0.423	0.437
8.5	0.3005	0.3171	0.3447	0.3752	0.4054	0.4325	0.4486	0.307	0.322	0.347	0.374	0.400	0.424	0.438
9	0.3022	0.3185	0.3458	0.3761	0.4064	0.4337	0.4500	0.308	0.323	0.348	0.375	0.402	0.425	0.439
9.5	0.3038	0.3199	0.3468	0.3770	0.4073	0.4347	0.4512	0.309	0.324	0.349	0.376	0.402	0.426	0.440
10	0.3054	0.3212	0.3478	0.3778	0.4082	0.4358	0.4525	0.310	0.325	0.350	0.377	0.403	0.427	0.440
10.5	0.3068	0.3224	0.3489	0.3788	0.4091	0.4369	0.4537	0.310	0.326	0.350	0.377	0.404	0.427	0.441
11	0.3082	0.3237	0.3500	0.3798	0.4102	0.4382	0.4551	0.311	0.326	0.351	0.378	0.405	0.428	0.442
11.5	0.3096	0.3250	0.3512	0.3810	0.4116	0.4397	0.4567	0.311	0.326	0.351	0.379	0.406	0.430	0.444
12	0.3111	0.3264	0.3526	0.3825	0.4132	0.4415	0.4587	0.311	0.326	0.352	0.380	0.407	0.432	0.446
12.5	0.3126	0.3280	0.3542	0.3843	0.4151	0.4436	0.4609	0.311	0.327	0.352	0.381	0.409	0.434	0.448
13	0.3140	0.3296	0.3561	0.3863	0.4174	0.4460	0.4634	0.312	0.327	0.353	0.382	0.410	0.436	0.451
13.5	0.3155	0.3312	0.3580	0.3885	0.4198	0.4486	0.4661	0.312	0.328	0.354	0.383	0.412	0.438	0.454
14	0.3168	0.3328	0.3600	0.3909	0.4224	0.4512	0.4687	0.313	0.328	0.354	0.384	0.413	0.440	0.457
14.5	0.3180	0.3344	0.3621	0.3933	0.4250	0.4538	0.4713	0.313	0.329	0.355	0.385	0.415	0.442	0.459
15	0.3191	0.3359	0.3640	0.3956	0.4275	0.4564	0.4738	0.314	0.329	0.355	0.385	0.416	0.444	0.461
15.5	0.3202	0.3373	0.3660	0.3979	0.4299	0.4587	0.4760	0.314	0.330	0.356	0.386	0.417	0.445	0.462
16	0.3212	0.3387	0.3678	0.4000	0.4322	0.4610	0.4781	0.315	0.330	0.356	0.387	0.418	0.446	0.464
16.5	0.3221	0.3400	0.3696	0.4021	0.4343	0.4630	0.4801	0.315	0.331	0.357	0.387	0.418	0.447	0.465
17	0.3230	0.3412	0.3712	0.4040	0.4363	0.4650	0.4819	0.316	0.331	0.357	0.388	0.419	0.448	0.466
17.5	0.3237	0.3423	0.3728	0.4058	0.4382	0.4668	0.4837	0.316	0.332	0.358	0.388	0.419	0.449	0.467
18	0.3244	0.3434	0.3743	0.4076	0.4401	0.4686	0.4853	0.317	0.332	0.358	0.389	0.420	0.449	0.467
Height (cm)														
120	0.294	0.311	0.339	0.369	0.398	0.424	0.439	0.300	0.316	0.342	0.369	0.396	0.419	0.432
125	0.297	0.314	0.342	0.372	0.401	0.428	0.443	0.303	0.319	0.345	0.372	0.398	0.421	0.434
130	0.300	0.316	0.344	0.375	0.404	0.431	0.447	0.307	0.322	0.348	0.375	0.401	0.423	0.437
135	0.302	0.319	0.347	0.377	0.407	0.434	0.450	0.309	0.325	0.350	0.377	0.403	0.426	0.439
140	0.304	0.321	0.349	0.379	0.410	0.437	0.453	0.310	0.326	0.351	0.378	0.405	0.427	0.441
145	0.306	0.323	0.350	0.381	0.412	0.440	0.457	0.311	0.327	0.352	0.380	0.406	0.429	0.443
150	0.308	0.325	0.352	0.384	0.415	0.443	0.460	0.311	0.327	0.353	0.380	0.408	0.431	0.446
155	0.310	0.327	0.355	0.386	0.418	0.447	0.464	0.312	0.327	0.353	0.381	0.410	0.435	0.449
160	0.313	0.330	0.358	0.389	0.421	0.450	0.468	0.312	0.328	0.353	0.383	0.412	0.439	0.455
165	0.316	0.333	0.360	0.392	0.424	0.453	0.471	0.313	0.328	0.354	0.384	0.415	0.443	0.461
170	0.320	0.336	0.364	0.395	0.427	0.457	0.474	0.313	0.328	0.354	0.385	0.417	0.448	0.466
175	0.323	0.339	0.367	0.398	0.430	0.459	0.477	0.314	0.329	0.355	0.386	0.420	0.452	0.472
180	0.326	0.342	0.370	0.401	0.433	0.462	0.480	0.315	0.329	0.356	0.387	0.422	0.456	0.478
185	0.329	0.345	0.372	0.403	0.435	0.464	0.482							
190	0.331	0.347	0.375	0.405	0.437	0.466	0.484							

Table S3: Percentile tables for Einc for boys and girls.

<i>EINC</i>														
Age and Height	<i>BOYS</i>							<i>GIRLS</i>						
	P5	P10	P25	P50	P75	P90	P95	P5	P10	P25	P50	P75	P90	P95
Age (years)														
6	0.0976	0.1098	0.1339	0.1678	0.2111	0.2608	0.2965	0.093	0.110	0.143	0.185	0.233	0.280	0.311
6.5	0.0996	0.1120	0.1366	0.1709	0.2148	0.2648	0.3006	0.095	0.113	0.145	0.186	0.233	0.280	0.311
7	0.1016	0.1142	0.1392	0.1740	0.2183	0.2687	0.3047	0.098	0.115	0.147	0.187	0.233	0.280	0.310
7.5	0.1036	0.1164	0.1418	0.1771	0.2218	0.2725	0.3085	0.100	0.117	0.148	0.188	0.234	0.280	0.309
8	0.1055	0.1186	0.1444	0.1801	0.2252	0.2760	0.3121	0.102	0.119	0.150	0.190	0.234	0.279	0.308
8.5	0.1076	0.1209	0.1470	0.1831	0.2285	0.2794	0.3153	0.105	0.122	0.152	0.191	0.235	0.279	0.308
9	0.1096	0.1232	0.1497	0.1861	0.2316	0.2824	0.3181	0.108	0.124	0.154	0.192	0.235	0.279	0.308
9.5	0.1118	0.1255	0.1523	0.1890	0.2346	0.2852	0.3206	0.112	0.127	0.156	0.193	0.236	0.279	0.308
10	0.1141	0.1280	0.1550	0.1918	0.2374	0.2876	0.3227	0.115	0.130	0.158	0.194	0.236	0.280	0.308
10.5	0.1166	0.1305	0.1577	0.1945	0.2399	0.2899	0.3246	0.118	0.132	0.160	0.195	0.237	0.281	0.310
11	0.1192	0.1332	0.1604	0.1972	0.2424	0.2918	0.3261	0.121	0.135	0.162	0.197	0.238	0.282	0.311
11.5	0.1217	0.1358	0.1631	0.1997	0.2446	0.2935	0.3273	0.123	0.137	0.163	0.198	0.239	0.283	0.313
12	0.1244	0.1385	0.1657	0.2022	0.2467	0.2950	0.3282	0.125	0.139	0.165	0.199	0.240	0.285	0.316
12.5	0.1270	0.1412	0.1684	0.2047	0.2489	0.2965	0.3293	0.127	0.140	0.166	0.200	0.242	0.287	0.319
13	0.1298	0.1440	0.1711	0.2073	0.2511	0.2982	0.3305	0.128	0.142	0.167	0.201	0.243	0.290	0.322
13.5	0.1327	0.1469	0.1740	0.2100	0.2534	0.3002	0.3321	0.130	0.143	0.168	0.202	0.245	0.292	0.324
14	0.1357	0.1499	0.1769	0.2128	0.2559	0.3023	0.3340	0.131	0.144	0.169	0.204	0.246	0.293	0.326
14.5	0.1389	0.1531	0.1800	0.2157	0.2586	0.3047	0.3361	0.131	0.145	0.170	0.205	0.247	0.294	0.327
15	0.1422	0.1564	0.1832	0.2188	0.2615	0.3073	0.3386	0.132	0.146	0.171	0.206	0.248	0.295	0.328
15.5	0.1457	0.1598	0.1866	0.2220	0.2646	0.3102	0.3414	0.133	0.146	0.172	0.207	0.249	0.295	0.327
16	0.1492	0.1633	0.1901	0.2255	0.2679	0.3134	0.3445	0.134	0.147	0.173	0.208	0.250	0.295	0.326
16.5	0.1528	0.1669	0.1937	0.2290	0.2713	0.3166	0.3476	0.134	0.148	0.175	0.209	0.251	0.295	0.325
17	0.1564	0.1706	0.1974	0.2326	0.2748	0.3199	0.3507	0.135	0.149	0.176	0.211	0.252	0.295	0.323
17.5	0.1601	0.1743	0.2011	0.2363	0.2783	0.3232	0.3538	0.136	0.150	0.177	0.212	0.252	0.294	0.322
18	0.1638	0.1780	0.2048	0.2399	0.2818	0.3265	0.3569	0.136	0.151	0.178	0.213	0.253	0.294	0.321
Height (cm)														
120	0.099	0.112	0.139	0.177	0.224	0.278	0.317	0.093	0.111	0.143	0.185	0.232	0.280	0.311
125	0.103	0.116	0.144	0.181	0.229	0.283	0.320	0.098	0.115	0.146	0.186	0.231	0.276	0.305
130	0.107	0.121	0.148	0.186	0.233	0.286	0.324	0.103	0.120	0.150	0.188	0.232	0.275	0.302
135	0.112	0.126	0.153	0.190	0.237	0.290	0.327	0.109	0.125	0.154	0.191	0.233	0.275	0.303
140	0.117	0.131	0.158	0.195	0.241	0.293	0.329	0.116	0.131	0.159	0.195	0.236	0.278	0.306
145	0.123	0.136	0.163	0.199	0.244	0.295	0.331	0.122	0.136	0.163	0.198	0.239	0.282	0.310
150	0.128	0.141	0.167	0.203	0.247	0.297	0.332	0.126	0.140	0.166	0.200	0.242	0.285	0.315
155	0.133	0.146	0.171	0.207	0.250	0.299	0.333	0.129	0.142	0.168	0.202	0.244	0.289	0.320
160	0.136	0.150	0.175	0.210	0.253	0.301	0.335	0.130	0.144	0.169	0.203	0.245	0.291	0.323
165	0.140	0.153	0.179	0.214	0.257	0.304	0.337	0.132	0.145	0.170	0.205	0.247	0.294	0.326
170	0.142	0.156	0.182	0.217	0.260	0.307	0.339	0.133	0.147	0.172	0.207	0.250	0.297	0.331
175	0.144	0.158	0.185	0.221	0.263	0.309	0.340	0.135	0.148	0.174	0.209	0.253	0.302	0.336
180	0.145	0.160	0.188	0.224	0.266	0.310	0.340	0.137	0.150	0.176	0.211	0.256	0.306	0.342
185	0.145	0.161	0.190	0.227	0.269	0.311	0.339							
190	0.145	0.161	0.192	0.229	0.271	0.313	0.339							

Table S4: Percentile tables for beta for boys and girls.

<i>beta</i>														
Age and Height	<i>BOYS</i>							<i>GIRLS</i>						
	P5	P10	P25	P50	P75	P90	P95	P5	P10	P25	P50	P75	P90	P95
Age (years)														
6	1.958	2.114	2.430	2.895	3.547	4.400	5.106	1.561	1.829	2.339	3.001	3.769	4.556	5.073
6.5	1.967	2.127	2.448	2.917	3.566	4.397	5.071	1.602	1.867	2.368	3.020	3.777	4.554	5.066
7	1.976	2.139	2.466	2.940	3.586	4.397	5.041	1.644	1.904	2.397	3.038	3.784	4.551	5.057
7.5	1.984	2.151	2.484	2.962	3.606	4.399	5.017	1.686	1.942	2.427	3.056	3.790	4.547	5.047
8	1.993	2.163	2.502	2.985	3.626	4.403	4.997	1.731	1.981	2.456	3.074	3.796	4.543	5.038
8.5	2.001	2.175	2.520	3.007	3.646	4.408	4.982	1.778	2.023	2.486	3.092	3.801	4.540	5.030
9	2.011	2.188	2.539	3.030	3.667	4.415	4.970	1.829	2.066	2.517	3.108	3.806	4.536	5.025
9.5	2.022	2.203	2.558	3.052	3.687	4.423	4.963	1.881	2.110	2.547	3.123	3.808	4.533	5.021
10	2.035	2.219	2.578	3.075	3.707	4.432	4.959	1.933	2.154	2.576	3.136	3.810	4.530	5.021
10.5	2.052	2.238	2.600	3.097	3.726	4.442	4.958	1.984	2.196	2.604	3.148	3.811	4.530	5.026
11	2.071	2.258	2.622	3.120	3.745	4.451	4.956	2.031	2.234	2.628	3.158	3.812	4.532	5.035
11.5	2.092	2.280	2.645	3.142	3.762	4.457	4.951	2.070	2.266	2.647	3.165	3.810	4.533	5.044
12	2.114	2.303	2.669	3.164	3.778	4.460	4.941	2.102	2.291	2.661	3.168	3.807	4.532	5.053
12.5	2.137	2.328	2.694	3.187	3.792	4.460	4.928	2.125	2.310	2.670	3.168	3.801	4.528	5.057
13	2.164	2.355	2.721	3.209	3.806	4.458	4.913	2.142	2.323	2.676	3.165	3.791	4.517	5.050
13.5	2.193	2.384	2.748	3.232	3.818	4.456	4.897	2.154	2.331	2.678	3.159	3.776	4.495	5.025
14	2.226	2.416	2.777	3.254	3.830	4.453	4.881	2.163	2.337	2.678	3.149	3.755	4.461	4.981
14.5	2.262	2.450	2.807	3.277	3.841	4.449	4.867	2.171	2.342	2.676	3.138	3.730	4.416	4.920
15	2.302	2.487	2.838	3.299	3.852	4.445	4.852	2.177	2.346	2.674	3.126	3.700	4.362	4.847
15.5	2.344	2.526	2.870	3.322	3.861	4.440	4.836	2.184	2.350	2.672	3.113	3.669	4.305	4.766
16	2.388	2.567	2.903	3.344	3.870	4.433	4.819	2.192	2.356	2.672	3.101	3.638	4.246	4.682
16.5	2.434	2.609	2.937	3.367	3.878	4.425	4.799	2.201	2.363	2.674	3.092	3.610	4.188	4.598
17	2.482	2.652	2.972	3.389	3.885	4.416	4.779	2.212	2.372	2.677	3.085	3.583	4.132	4.518
17.5	2.530	2.696	3.007	3.411	3.893	4.407	4.759	2.224	2.382	2.683	3.080	3.559	4.080	4.442
18	2.578	2.739	3.041	3.434	3.900	4.398	4.738	2.237	2.394	2.689	3.075	3.536	4.031	4.371
Height (cm)														
120	1.940	2.124	2.488	3.001	3.668	4.455	5.039	1.597	1.898	2.462	3.182	4.001	4.824	5.356
125	1.963	2.148	2.513	3.024	3.685	4.459	5.029	1.679	1.963	2.495	3.173	3.946	4.727	5.233
130	1.992	2.178	2.543	3.051	3.704	4.462	5.018	1.766	2.031	2.527	3.164	3.896	4.642	5.129
135	2.031	2.216	2.578	3.080	3.723	4.466	5.007	1.859	2.102	2.560	3.155	3.848	4.567	5.042
140	2.081	2.263	2.619	3.112	3.742	4.470	5.001	1.952	2.172	2.593	3.146	3.805	4.503	4.974
145	2.136	2.314	2.662	3.143	3.758	4.470	4.991	2.031	2.233	2.619	3.137	3.767	4.453	4.928
150	2.190	2.364	2.704	3.173	3.773	4.468	4.977	2.092	2.278	2.638	3.128	3.737	4.420	4.904
155	2.238	2.410	2.744	3.204	3.790	4.465	4.960	2.134	2.309	2.650	3.119	3.714	4.398	4.895
160	2.281	2.451	2.782	3.235	3.807	4.462	4.937	2.168	2.334	2.658	3.110	3.693	4.379	4.889
165	2.315	2.487	2.817	3.264	3.822	4.451	4.902	2.203	2.360	2.668	3.101	3.669	4.353	4.875
170	2.337	2.511	2.844	3.288	3.833	4.433	4.854	2.238	2.386	2.678	3.091	3.643	4.322	4.853
175	2.340	2.520	2.859	3.304	3.837	4.409	4.801	2.271	2.411	2.688	3.082	3.616	4.288	4.826
180	2.324	2.511	2.861	3.310	3.835	4.382	4.749	2.302	2.434	2.697	3.073	3.588	4.250	4.793
185	2.289	2.487	2.850	3.307	3.827	4.354	4.699							
190	2.240	2.451	2.831	3.299	3.816	4.325	4.650							

Table S5: Percentile tables for DC for boys and girls.

DC														
Age and Height	BOYS							GIRLS						
	P5	P10	P25	P50	P75	P90	P95	P5	P10	P25	P50	P75	P90	P95
Age (years)														
6	32.00	40.36	54.31	69.80	85.26	99.17	107.49	36.79	42.68	53.42	66.65	81.23	95.48	104.53
6.5	32.71	40.57	53.91	68.96	84.22	98.09	106.45	36.96	42.67	53.12	66.05	80.38	94.47	103.44
7	33.32	40.73	53.50	68.14	83.18	97.02	105.42	37.11	42.65	52.81	65.46	79.54	93.45	102.35
7.5	33.84	40.85	53.08	67.32	82.15	95.95	104.40	37.25	42.62	52.51	64.87	78.70	92.43	101.24
8	34.27	40.92	52.65	66.50	81.12	94.88	103.37	37.36	42.58	52.21	64.29	77.86	91.40	100.12
8.5	34.64	40.95	52.22	65.69	80.09	93.79	102.31	37.44	42.53	51.92	63.73	77.04	90.35	98.94
9	34.92	40.95	51.79	64.89	79.06	92.66	101.18	37.49	42.46	51.64	63.19	76.22	89.27	97.71
9.5	35.15	40.92	51.36	64.11	78.01	91.48	99.96	37.51	42.38	51.36	62.66	75.40	88.17	96.43
10	35.33	40.87	50.96	63.35	76.96	90.24	98.64	37.48	42.26	51.08	62.14	74.60	87.08	95.13
10.5	35.47	40.82	50.57	62.60	75.89	88.92	97.21	37.37	42.09	50.76	61.62	73.82	86.01	93.86
11	35.61	40.77	50.20	61.87	74.82	87.56	95.69	37.20	41.87	50.43	61.12	73.08	84.99	92.65
11.5	35.77	40.75	49.86	61.17	73.76	86.19	94.13	36.99	41.62	50.10	60.64	72.40	84.07	91.55
12	35.92	40.72	49.53	60.48	72.69	84.79	92.55	36.75	41.35	49.77	60.19	71.78	83.24	90.57
12.5	36.05	40.68	49.18	59.76	71.59	83.35	90.89	36.51	41.10	49.46	59.78	71.22	82.50	89.70
13	36.12	40.59	48.79	59.00	70.43	81.80	89.11	36.34	40.90	49.19	59.40	70.70	81.81	88.90
13.5	36.15	40.46	48.36	58.19	69.19	80.14	87.17	36.29	40.80	48.99	59.07	70.22	81.18	88.16
14	36.11	40.27	47.88	57.32	67.87	78.35	85.08	36.37	40.81	48.87	58.80	69.77	80.56	87.44
14.5	36.03	40.04	47.34	56.39	66.47	76.47	82.88	36.58	40.93	48.84	58.58	69.35	79.95	86.71
15	35.90	39.75	46.75	55.40	65.00	74.49	80.58	36.91	41.16	48.88	58.40	68.94	79.33	85.97
15.5	35.73	39.42	46.10	54.33	63.44	72.43	78.19	37.34	41.47	48.97	58.24	68.53	78.70	85.21
16	35.52	39.03	45.38	53.19	61.81	70.29	75.72	37.83	41.82	49.08	58.07	68.09	78.02	84.40
16.5	35.27	38.60	44.62	51.99	60.11	68.10	73.19	38.35	42.19	49.19	57.88	67.60	77.29	83.53
17	34.99	38.15	43.83	50.77	58.40	65.89	70.66	38.88	42.56	49.29	57.67	67.07	76.48	82.57
17.5	34.69	37.67	43.03	49.55	56.72	63.73	68.19	39.39	42.91	49.36	57.42	66.50	75.62	81.54
18	34.36	37.17	42.22	48.35	55.06	61.62	65.78	39.88	43.25	49.43	57.17	65.91	74.73	80.48
Height (cm)														
120	33.07	40.07	52.54	67.41	83.25	98.26	107.56	36.15	41.47	51.47	64.31	79.10	94.20	104.08
125	33.80	40.34	52.10	66.31	81.65	96.35	105.53	36.84	41.93	51.48	63.71	77.79	92.17	101.58
130	34.35	40.48	51.59	65.15	79.92	94.21	103.20	37.35	42.23	51.36	63.03	76.46	90.16	99.14
135	34.69	40.49	51.02	63.92	78.03	91.74	100.39	37.68	42.37	51.12	62.29	75.11	88.19	96.76
140	34.86	40.39	50.40	62.64	76.00	88.97	97.16	37.78	42.32	50.77	61.51	73.79	86.28	94.45
145	34.97	40.25	49.77	61.35	73.94	86.13	93.80	37.71	42.15	50.37	60.76	72.57	84.50	92.27
150	35.08	40.10	49.12	60.06	71.93	83.40	90.61	37.50	41.89	49.97	60.10	71.52	82.97	90.37
155	35.16	39.90	48.43	58.77	70.00	80.85	87.67	37.29	41.67	49.68	59.65	70.78	81.85	88.96
160	35.21	39.67	47.71	57.47	68.11	78.43	84.95	37.12	41.52	49.52	59.38	70.28	81.00	87.85
165	35.32	39.48	47.00	56.20	66.30	76.17	82.44	36.91	41.36	49.36	59.09	69.70	80.00	86.50
170	35.52	39.37	46.37	55.04	64.68	74.23	80.36	36.57	41.08	49.09	58.67	68.95	78.80	84.95
175	35.78	39.32	45.83	54.03	63.32	72.70	78.82	36.14	40.70	48.71	58.14	68.09	77.47	83.26
180	36.05	39.31	45.40	53.19	62.24	71.60	77.82	35.66	40.29	48.30	57.56	67.18	76.11	81.57
185	36.32	39.35	45.06	52.53	61.42	70.89	77.34							
190	36.57	39.39	44.77	51.96	60.76	70.44	77.22							

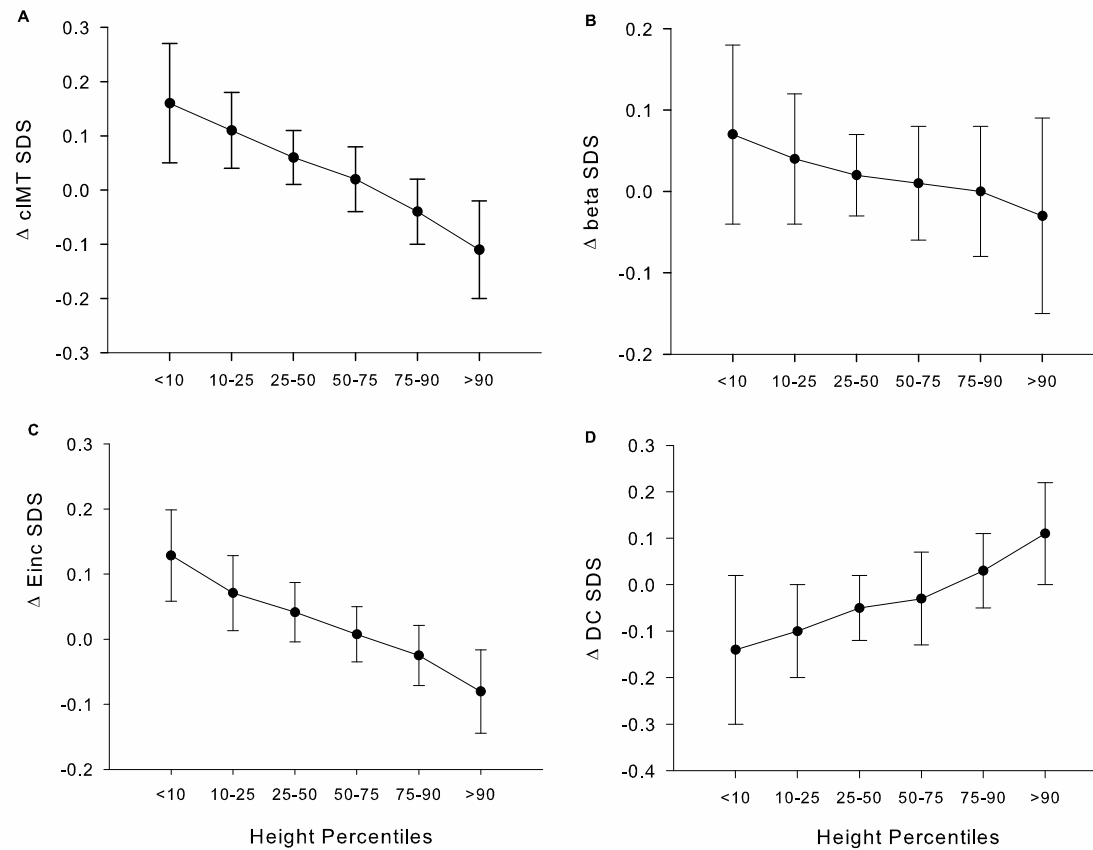
Table S6: LMS tables for cIMT, Einc, Beta and DC for boys.

BOYS	cIMT			Einc			Beta			DC		
	L	M	S	L	M	S	L	M	S	L	M	S
Age (years)												
6	1.444	0.370	0.121	-0.091	0.168	0.337	-0.786	2.895	0.278	1.005	69.80	0.329
6.5	1.377	0.371	0.121	-0.082	0.171	0.336	-0.721	2.917	0.277	0.940	68.97	0.326
7	1.309	0.372	0.121	-0.073	0.174	0.334	-0.657	2.940	0.276	0.876	68.14	0.323
7.5	1.239	0.373	0.121	-0.063	0.177	0.332	-0.594	2.962	0.275	0.811	67.32	0.320
8	1.165	0.374	0.120	-0.051	0.180	0.329	-0.532	2.985	0.274	0.748	66.50	0.318
8.5	1.086	0.375	0.120	-0.038	0.183	0.327	-0.472	3.007	0.273	0.687	65.70	0.315
9	1.004	0.376	0.119	-0.025	0.186	0.324	-0.417	3.030	0.272	0.632	64.90	0.312
9.5	0.925	0.377	0.119	-0.013	0.189	0.320	-0.370	3.052	0.270	0.583	64.11	0.308
10	0.851	0.378	0.118	-0.005	0.192	0.316	-0.332	3.075	0.269	0.542	63.34	0.304
10.5	0.787	0.379	0.118	-0.001	0.195	0.311	-0.303	3.097	0.267	0.509	62.60	0.300
11	0.735	0.380	0.118	0.001	0.197	0.306	-0.281	3.120	0.264	0.481	61.88	0.295
11.5	0.694	0.381	0.117	0.005	0.200	0.301	-0.258	3.142	0.261	0.455	61.17	0.289
12	0.667	0.382	0.117	0.008	0.202	0.295	-0.234	3.164	0.257	0.429	60.48	0.284
12.5	0.655	0.384	0.117	0.010	0.205	0.290	-0.209	3.187	0.253	0.405	59.76	0.278
13	0.664	0.386	0.118	0.008	0.207	0.284	-0.188	3.209	0.249	0.386	59.00	0.272
13.5	0.695	0.389	0.118	0.002	0.210	0.279	-0.173	3.232	0.244	0.374	58.19	0.265
14	0.746	0.391	0.118	-0.007	0.213	0.274	-0.167	3.254	0.238	0.367	57.32	0.258
14.5	0.813	0.393	0.119	-0.020	0.216	0.269	-0.172	3.277	0.233	0.364	56.39	0.251
15	0.890	0.396	0.119	-0.033	0.219	0.264	-0.185	3.299	0.226	0.363	55.40	0.244
15.5	0.970	0.398	0.119	-0.047	0.222	0.259	-0.206	3.322	0.220	0.359	54.33	0.236
16	1.048	0.400	0.119	-0.061	0.225	0.254	-0.232	3.344	0.213	0.353	53.19	0.229
16.5	1.124	0.402	0.119	-0.073	0.229	0.250	-0.264	3.367	0.206	0.346	51.99	0.221
17	1.198	0.404	0.119	-0.084	0.233	0.245	-0.300	3.389	0.199	0.337	50.77	0.213
17.5	1.271	0.406	0.119	-0.094	0.236	0.241	-0.340	3.411	0.191	0.328	49.56	0.205
18	1.342	0.408	0.120	-0.102	0.240	0.237	-0.382	3.434	0.184	0.318	48.35	0.197
Height (cm)												
120	1.274	0.369	0.119	0.000	0.177	0.355	-0.363	3.001	0.287	0.725	67.41	0.338
125	1.234	0.372	0.119	-0.003	0.181	0.346	-0.349	3.024	0.283	0.660	66.31	0.331
130	1.188	0.375	0.119	-0.011	0.186	0.337	-0.337	3.051	0.279	0.606	65.15	0.323
135	1.133	0.377	0.119	-0.030	0.190	0.326	-0.342	3.080	0.272	0.574	63.92	0.313
140	1.076	0.379	0.120	-0.065	0.195	0.314	-0.375	3.112	0.264	0.568	62.64	0.303
145	1.020	0.381	0.120	-0.104	0.199	0.301	-0.424	3.143	0.255	0.572	61.35	0.292
150	0.966	0.384	0.121	-0.138	0.203	0.289	-0.470	3.173	0.247	0.569	60.06	0.282
155	0.908	0.386	0.121	-0.162	0.207	0.280	-0.497	3.204	0.239	0.551	58.77	0.272
160	0.840	0.389	0.121	-0.173	0.210	0.273	-0.495	3.235	0.232	0.515	57.47	0.263
165	0.774	0.392	0.120	-0.160	0.214	0.267	-0.453	3.264	0.226	0.455	56.20	0.254
170	0.714	0.395	0.119	-0.111	0.217	0.263	-0.359	3.288	0.221	0.357	55.04	0.246
175	0.670	0.398	0.118	-0.023	0.221	0.261	-0.222	3.304	0.218	0.222	54.03	0.239
180	0.650	0.401	0.117	0.097	0.224	0.259	-0.056	3.310	0.217	0.057	53.20	0.234
185	0.662	0.403	0.115	0.237	0.227	0.257	0.130	3.307	0.218	-0.127	52.53	0.230
190	0.694	0.405	0.114	0.388	0.229	0.257	0.329	3.299	0.221	-0.323	51.96	0.226

Table S7: LMS tables for cIMT, Einc, Beta and DC for girls.

GIRLS	cIMT			Einc			Beta			DC		
	L	M	S	L	M	S	L	M	S	L	M	S
Age (years)												
6	1.268	0.366	0.112	0.457	0.185	0.358	0.373	3.001	0.353	0.535	66.65	0.309
6.5	1.262	0.368	0.111	0.452	0.186	0.351	0.354	3.020	0.345	0.504	66.05	0.306
7	1.256	0.369	0.109	0.447	0.187	0.344	0.334	3.038	0.338	0.473	65.46	0.303
7.5	1.251	0.371	0.108	0.441	0.188	0.336	0.312	3.056	0.330	0.442	64.87	0.299
8	1.250	0.373	0.107	0.428	0.190	0.329	0.284	3.074	0.322	0.414	64.29	0.296
8.5	1.254	0.374	0.106	0.405	0.191	0.321	0.246	3.092	0.314	0.392	63.73	0.292
9	1.262	0.375	0.105	0.370	0.192	0.313	0.195	3.108	0.306	0.377	63.19	0.288
9.5	1.273	0.376	0.105	0.322	0.193	0.306	0.133	3.123	0.298	0.370	62.66	0.284
10	1.282	0.377	0.105	0.263	0.194	0.299	0.059	3.136	0.290	0.369	62.14	0.280
10.5	1.283	0.377	0.105	0.194	0.195	0.293	-0.028	3.148	0.283	0.376	61.62	0.277
11	1.272	0.378	0.106	0.122	0.197	0.288	-0.119	3.158	0.276	0.388	61.12	0.275
11.5	1.244	0.379	0.107	0.053	0.198	0.284	-0.210	3.165	0.270	0.404	60.64	0.272
12	1.198	0.380	0.108	-0.008	0.199	0.281	-0.294	3.168	0.265	0.420	60.19	0.271
12.5	1.133	0.381	0.110	-0.060	0.200	0.280	-0.366	3.168	0.261	0.435	59.78	0.270
13	1.053	0.382	0.111	-0.100	0.201	0.279	-0.422	3.165	0.258	0.444	59.40	0.268
13.5	0.963	0.383	0.113	-0.125	0.202	0.279	-0.459	3.159	0.254	0.444	59.08	0.266
14	0.873	0.384	0.114	-0.135	0.204	0.278	-0.479	3.149	0.250	0.437	58.80	0.263
14.5	0.793	0.385	0.115	-0.130	0.205	0.277	-0.488	3.138	0.245	0.424	58.58	0.259
15	0.729	0.385	0.116	-0.109	0.206	0.276	-0.485	3.126	0.240	0.404	58.40	0.255
15.5	0.680	0.386	0.117	-0.075	0.207	0.274	-0.473	3.113	0.235	0.376	58.24	0.249
16	0.639	0.387	0.117	-0.030	0.208	0.271	-0.451	3.101	0.229	0.340	58.07	0.242
16.5	0.603	0.387	0.118	0.023	0.209	0.268	-0.420	3.092	0.222	0.297	57.88	0.236
17	0.569	0.388	0.118	0.081	0.211	0.266	-0.384	3.085	0.216	0.249	57.67	0.228
17.5	0.534	0.388	0.118	0.142	0.212	0.263	-0.343	3.080	0.209	0.199	57.42	0.221
18	0.591	0.391	0.117	0.204	0.213	0.260	-0.301	3.075	0.203	0.148	57.17	0.213
Height (cm)												
120	1.585	0.369	0.108	0.449	0.185	0.357	0.467	3.182	0.358	0.340	64.31	0.318
125	1.588	0.372	0.106	0.469	0.186	0.338	0.425	3.173	0.339	0.317	63.71	0.306
130	1.586	0.375	0.105	0.453	0.188	0.321	0.353	3.164	0.320	0.296	63.03	0.295
135	1.571	0.377	0.104	0.392	0.191	0.306	0.241	3.155	0.302	0.281	62.29	0.285
140	1.533	0.378	0.104	0.290	0.195	0.293	0.088	3.146	0.284	0.279	61.51	0.277
145	1.462	0.380	0.105	0.167	0.198	0.284	-0.087	3.137	0.269	0.297	60.76	0.270
150	1.338	0.380	0.107	0.050	0.200	0.279	-0.261	3.128	0.258	0.330	60.10	0.265
155	1.137	0.381	0.110	-0.038	0.202	0.276	-0.418	3.119	0.250	0.372	59.65	0.262
160	0.872	0.383	0.113	-0.091	0.203	0.275	-0.561	3.110	0.243	0.423	59.38	0.259
165	0.604	0.384	0.117	-0.129	0.205	0.275	-0.712	3.101	0.235	0.496	59.09	0.255
170	0.378	0.385	0.121	-0.161	0.207	0.276	-0.870	3.091	0.227	0.583	58.67	0.251
175	0.169	0.386	0.124	-0.193	0.209	0.277	-1.030	3.082	0.218	0.677	58.14	0.247
180	-0.043	0.387	0.127	-0.225	0.211	0.277	-1.192	3.073	0.210	0.774	57.56	0.243

Figure S1: Difference between height- and age-related SDS of CIMT (A), beta (B), Einc (C) and DC (D) according to height percentiles. Data are given as mean $\pm$ SEM.



Equations.

Distensibility coefficient

$$DC \text{ (kPa}^{-1} \cdot 10^{-3}) = 2 \cdot (\Delta D / DD) / (\Delta P \cdot 0.1333) \cdot 1000$$

Stiffness index

$$\beta = \ln(SBP/DBP) / (\Delta D / DD)$$

Incremental modulus of elasticity:

$$E_{inc} \text{ (kPa} \cdot 10^3) = 3 \cdot (1 + LCSA / IMCSA) / DC$$

SD = mean systolic carotid diameter

DD = mean diastolic carotid diameter

SBP = systolic blood pressure

DBP = diastolic blood pressure

MAP = mean arterial blood pressure

cIMT = carotid intima-media thickness

$$\Delta D = SD - DD$$

$$\Delta P = SBP - DBP$$

$$LCSA = 3.14 \cdot DD^2 / 4$$

$$IMCSA = 3.14 \cdot (DD/2 + cIMT)^2 - 3.14 \cdot (DD/2)^2$$